

ARNAV VERMA

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PROFESSIONAL SUMMARY

Embedded Systems Engineer specialising in **embedded hardware, firmware, and autonomous systems**.

Hands-on experience designing **KiCad PCBs, power electronics, and embedded C/C++ firmware**, alongside building and flight-testing **autonomous UAV platforms** using **PX4, ArduPilot, MAVLink, ROS 2, and companion computers**. Experienced in bridging hardware and software from **schematic and PCB design to firmware, system integration, and real-world deployment**, with additional work in **wireless communications and robotic systems**.

Education

Thapar Institute of Engineering and Technology, BE in Electronics and Computer Aug 2023 – Aug 2027

- CGPA: **9.34/10.0** | Digital System Design, Analog Electronics, Computer Architecture, Embedded Systems, DSP

Work Experience

UAV Systems Intern, EME corps, Indian Army March 2026

- Completed a 3-week internship with the EME Corps, contributing to a Hailing Swarm Drone System
- Awarded a **Certificate of Internship** by EME for successful project delivery

Embedded Systems and UAV Engineer, UPWORK, Freelance

April 2026 – Present

- Designed PCBs, embedded firmware, and Gazebo-based UAV simulations for international clients.
- Assembled and integrated UAV electronics and RF communication modules.

Head of Avionics, Team Sammpaati, TIET

August 2025 – Present

- Led a 10-member team for the NIDAR Disaster-Management Drone Competition (DFI), developing and integrating an autonomous UAV platform.
- Conducted UAV workshops and trained **40+ juniors** in embedded systems and communication protocols

Projects

Fully Autonomous Heavy-Lift UAV for Disaster Response Flight Video

- Designed, manufactured, and flight-tested a custom coaxial UAV with a **1000mm wheelbase**, hybrid 3D-printed and carbon-fiber tube airframe, using **18" rotors** to achieve **22 min endurance** and **5 kg payload capacity**
- Integrated **Pixhawk + Raspberry Pi** companion computer; built KML-driven autonomous missions (area scan, survivor detection, payload delivery) and an onboard **OpenCV + ML** inference pipeline for real-time flood-survivor detection
- Enabled **4G/LTE** based long-range communication and autonomous payload release

Universal Isolated SMPS PCB

- Designed a **30W**, universal-input (**85–265 VAC**) offline flyback SMPS in **KiCad 9** around a **TOP244Y** controller with custom transformer footprint, **4kV** isolation, EMI/surge protection, and isolated feedback; produced manufacturing-ready Gerbers with dual **12V/5V** outputs.

Custom FHSS Radio Link for UAV Swarms

- Built a custom **ESP32 + SX1278 LoRa FHSS** communication system for reliable **UAV swarm** telemetry.
- Implemented **frequency hopping**, synchronization, and embedded communication protocols; verified RF performance with **HackRF One**.

Reefer Container Drone Inspection System

- Developed an autonomous UAV inspection system with a **BAPLIE D95B EDI** parser that automatically generates photorealistic **Gazebo** simulation worlds from vessel stow plans.
- Built a **PX4 SITL** offboard controller using **MAVSDK** for automated container inspection, with a Python **Tkinter** GUI featuring live mapping and CSV logging.

Skills

Languages: C, C++, Python

Embedded Systems: STM32, ESP32, Arduino, UART, SPI, I2C, PWM, CAN, RS485

Companion Computers: Raspberry Pi 4/5, Linux (Ubuntu), Shell scripting, Jetson Nano, JetBot

Drone Autonomy and Control: PX4, Ardupilot, MAVLink/MAVSDK, DroneKit, SITL, Gazebo, Python

Communication and Networking: Wi-Fi, Bluetooth, LoRa, Adalm Pluto SDR, HackRF One, GNU Radio, GQRX

Tools and Platforms: KiCad, MAVProxy, QGroundControl, Mission Planner, PlatformIO